

UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY | SOFTWARE ENGINEERING DEPARTMENT

INTRODUCTION TO SOFTWARE ENGINEERING

PROJECT ASSIGNMENT 1

Project Proposal & Plan

HomeMate AI

AI-assisted coordination for self-managed residential construction

DOCUMENT INFORMATION	DETAILS
Team ID	T08
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Teaching Assistant

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Objectives

This Project Assignment 1 (PA1) document establishes the preliminary product vision, proposed solution and management baseline for Team T08's HomeMate AI project. It follows the course Project Proposal & Plan template and the approved team-contract principles from PA0.

- Define the problem, intended users, feature priorities and exclusions.
- Describe a feasible system context and technology approach.
- Establish WBS, effort estimates, schedule, responsibilities, costs, risks and collaboration procedures.
- Provide spaces for member contribution evidence and the AI usage log, to be completed by the team before submission.

1 Member Contribution Assessment

Contribution percentages and supporting evidence are intentionally left blank for the team lead to complete using PA1 work records. The final percentages must total 100%, with task-board, commit or pull-request evidence as applicable.

Student ID	Full name	Role(s)	Contribution (%)	Evidence
24127385	Huỳnh Minh Hùng	Team Lead / PM	50%	https://clc-team-dj4m9b5b.atlassian.net/browse/SCRUM-6
24127464	Nguyễn Phạm Minh Ngọc	Business Analyst	25%	https://clc-team-dj4m9b5b.atlassian.net/browse/SCRUM-5
24127515	Phạm Minh Quân	Designer	0%	
21127061	Trần Gia Huy	Developer	25%	https://clc-team-dj4m9b5b.atlassian.net/browse/SCRUM-7
24127570	Võ Thúc Trí	Tester	0%	
	TOTAL		100%	

To be completed by the team: contribution values, evidence links, and final verification of the 100% total.

2 Vision

2.1 Preliminary Problem Statement

Business context and problem

In Vietnam, many households choose to manage their own residential construction or renovation instead of hiring a full-service contractor. The homeowner directly purchases materials, hires workers, and monitors progress and workmanship. However, records and coordination frequently remain fragmented across notebooks, printed receipts and informal conversations.

Managing a household construction project requires the coordination of several interdependent concerns: the budget and material purchases, workforce attendance and wages, the order and timing of construction stages, and site evidence. For homeowners without specialist construction knowledge, scattered information makes it difficult to identify cost overruns early, recognize delayed activities, and maintain consistent records of project progress. General personal-expense applications and general-purpose AI chat tools do not necessarily organize information around the construction stages of a specific household project.

In a self-managed construction process, purchases, labour records and planned stages influence one another. A material purchase affects the remaining budget, attendance determines estimated labour payments, and missed stages may require changes to planned dates. When this information is maintained in separate records, even a basic question—such as how much has been spent on the current stage or which activities still require attention—may require repeated manual reconciliation. The proposed system therefore emphasizes a consistent, owner-controlled project record and straightforward summaries rather than complex professional construction analysis.

The primary user is the household owner, who provides initial project information and maintains costs, schedules and worker data. A crew leader may assist with attendance or assigned-stage updates only when the owner grants the corresponding permission. Because project inputs may be incomplete or imprecise, generated schedules must remain editable and financial figures must be presented as records or estimates rather than certified measurements. HomeMate AI aims to improve information organization and day-to-day coordination; responsibility for on-site safety, structural decisions and legal compliance remains with qualified professionals and the project participants.

HomeMate AI is proposed as a project-oriented responsive web application that brings cost records, worker attendance, an editable stage schedule, and context-aware AI assistance into one workflow. It is intended to support better-informed homeowner monitoring and preliminary planning, rather than perform professional engineering supervision or independent technical certification.

Operating environment

The application runs in modern HTML5-capable browsers on computers, tablets and mobile phones with an Internet connection. A cloud-hosted frontend and managed backend let homeowners review budget and schedule information from home, while authorized crew leaders can record attendance or stage progress at the construction site without installing native software.

Preliminary design and implementation constraints

- Web interface: React with Tailwind CSS and responsive layouts; native mobile applications are outside the initial commitment.
- Managed backend: Supabase provides PostgreSQL, authentication, Row Level Security, Edge Functions and optional file storage; the frontend is deployed through Vercel.
- AI integration: Google Gemini is accessed from server-side Supabase Edge Functions using minimized, bounded project context; credentials and sensitive values must not be exposed in the browser.
- Safety and legal limits: AI outputs are preliminary guidance, cannot certify structural safety or regulatory compliance, and must present clear disclaimers and recommend qualified professional verification.
- Course constraints: a five-member team follows the defined PA1–PA5 Waterfall phase gates; version-controlled documentation, reviewed pull requests, conventional commits and weekly reporting are required.

2.2 Product Position Statement

Element	Content
For	Vietnamese homeowners and families who independently manage home building or renovation, together with crew leaders granted limited access.
Who	Need a coherent way to coordinate budgets, workforce attendance, work sequencing, construction-stage information and site records instead of relying on fragmented notes.
The HomeMate AI	A web-based, AI-assisted coordination application for self-managed residential construction and renovation.
That	Combines material and labour expenditures, wage estimates, editable stage schedules and project-context questions in one homeowner-oriented workflow.
Unlike	Separate expense trackers (e.g., Money Lover), financial systems (e.g., MISA), general AI assistants or handwritten records.
Our product	Connects cost, progress and workforce records to relevant AI guidance while maintaining a simple interface and visible technical-safety limitations.

2.3 Stakeholders and Users

#	Stakeholder / user	Description	Main interests
1	Project sponsor / teaching staff	Course instructor and teaching assistant reviewing each PA baseline.	Feasible MVP, transparent teamwork, process compliance and quality deliverables.
2	Household owner / family	Primary end users self-managing construction or renovation.	Budget and material records; stage planning; attendance and wages; context-aware support; optional evidence.
3	Crew leader / coordinator	Secondary users invited and authorized by the owner.	Access assigned work, record attendance and progress within granted permissions.

#	Stakeholder / user	Description	Main interests
4	Team T08	Five students covering business analysis, design, development, QA and project management.	Deliver Must-have features, credible tests and PA milestones within the course schedule.
5	Managed service providers	Supabase and Google Gemini, with Vercel for web delivery.	Reliable service integration, data minimization, availability and quota compliance.

2.4 Scope and Key Features

MoSCoW priorities identify mandatory release scope. Only Must-have features are committed for full implementation in PA5; Should-have and Could-have functionality may be deferred when schedule or quality is at risk.

ID	Feature	Description	Priority
F01	Cost and material records	Budget categories; itemized materials and labour expenses; quantity, unit price, transaction date, calculated totals, dashboard and visual budget-threshold indicators.	Must
F02	AI-assisted construction-stage planning	Capture project type, approximate scale and preferred dates; request a draft stage sequence from Gemini; edit, reorder and save stages; update stage status.	Must
F03	Worker attendance and wage estimates	Maintain worker/crew profiles and rates; record full/half day or hourly attendance; estimate payable wages; review period totals.	Must
F04	Context-aware AI question answering	In-app chat backed by minimized structured project context; answer planning and record questions with explicit safety and legal disclaimers.	Must
F05	Stage checklists and reminders	Provide stage-specific checklists with reviewed sources and applicable limitations.	Should
F06	Construction evidence archive	Attach and manage site photos, purchase receipts and acceptance notes by date or stage.	Could
F07	Exception alerts and delay notifications	Proactive in-app alerts for threshold overruns or schedule delays; distinct from F01 dashboard color indicators.	Could
F08	Native mobile application	Separate installed Android/iOS application beyond responsive web access.	Won't

Out of scope: receipt OCR; online payments; professional structural calculation, certification or legal advice; detailed 2D/3D architectural modelling; interior design or feng shui optimization; materials marketplaces; biometric attendance; worker recruitment or payroll transfer; native mobile apps; and support for more than two concurrent construction projects in the initial release, as proposed by the BA scope draft.

Safety boundary: AI-generated information must be reviewed by the user and, when safety-critical, verified with qualified professionals. HomeMate AI is not a substitute for an architect, structural engineer, site supervisor or official building inspection.

3 Proposed Solution

3.1 Preliminary Architecture

HomeMate AI is treated as one software system accessed by a Household Owner and an authorized Crew Leader. External managed services support identity, data and AI features. This C4 Level 1 System Context view intentionally does not specify container-level design, which is deferred to PA3.

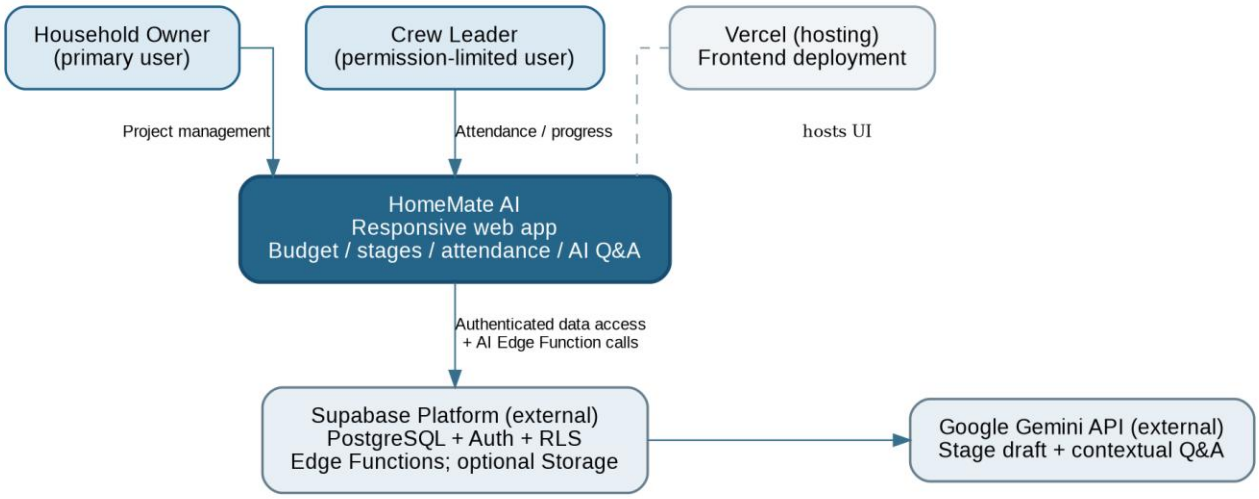


Figure 3.1 – Preliminary HomeMate AI System Context (C4 Level 1).

Element	Type	Responsibility
Household Owner	Person	Maintains project data, expenses, schedule, workers, permissions and AI questions.
Crew Leader	Person	Enters authorized attendance and stage progress; no broad budget access without a grant.
HomeMate AI	Software system	Responsive project-oriented web experience for F01–F04.
Supabase	External platform	Managed Auth, PostgreSQL, RLS, Edge Functions and optional Storage.
Google Gemini	External API	Provides bounded AI stage drafts and context-aware answers without direct database access.
Vercel	Hosting platform	Serves the React frontend and deployment previews.

The baseline uses a Supabase-centric Backend-as-a-Service architecture rather than a dedicated FastAPI/Node.js backend server. For routine operations, the React client uses the Supabase SDK with project-level RLS enforcement. For AI functions, the client invokes authenticated Supabase Edge Functions which validate permissions, prepare minimized context, call Gemini with server-held secrets and validate the results.

Key decisions: (AD-01) managed BaaS and serverless functions to minimize infrastructure overhead; (AD-02) authorization enforced with RLS and verified in tests; (AD-03) AI-generated stage plans remain editable drafts until owner confirmation; (AD-04) use structured SQL-based context summaries rather than vector RAG in the MVP; (AD-05) external AI input is minimized and safety disclaimers are mandatory. Detailed technical decisions may be revised and documented in PA3.

3.2 Technology Stack

Layer	Technology	Reason for choice
Front end	React + TypeScript; Vite; Tailwind CSS	Responsive web UI and maintainable typed components; development effort focused on homeowner workflows.
Back end	Supabase Auth, REST/data APIs, Edge Functions (TypeScript)	Managed authentication and CRUD; secure server-side AI calls without operating a separate server.
Database	Supabase PostgreSQL; constraints and RLS	Relational model fits project expenses, stages, workers and attendance; authorization enforced by database policies.
AI integration	Google Gemini API from Edge Functions	Draft stage plans and answer bounded project-context questions with validated structured output.
Hosting / deployment	Vercel frontend; Supabase Cloud	Simple hosted workflow; free-tier usage targeted subject to quotas and current platform policies.

Layer	Technology	Reason for choice
Testing & CI	GitHub Actions; lightweight frontend tests; RLS and Edge Function tests; selective end-to-end tests	Continuous checks for core flows and access rules; exact test tools selected during design/implementation.
Collaboration	GitHub; Jira; Microsoft Teams; Google Drive	Source control, work-package tracking, internal coordination and shared documents.

Optional libraries (for example Zod, React Testing Library or Playwright) may be introduced only when their value justifies the additional integration effort. The proposal deliberately excludes dedicated FastAPI, Render hosting and production Docker deployment from the baseline.

3.3 Hardware and Operating Environment

Environment	Hardware / service	Requirements and notes
Development	Student laptop or desktop; Windows, macOS or Linux	Node.js LTS, Git, IDE and browser; Supabase CLI when needed. An 8 GB RAM machine is sufficient for browser-centric development, with 16 GB desirable.
Testing	Desktop browser and representative smartphone browser; GitHub-hosted CI	Test responsive display, permissions, calculations, stage editing, AI error handling and representative network conditions.
Production / demo	Vercel + Supabase Cloud + Gemini API	HTTPS connectivity, authenticated browser clients, private server-side AI secrets and budgeted API usage.
End user	Modern browser on desktop, tablet or phone	Internet access required; no native installation or dedicated on-premises server.

The system uses existing student devices and managed cloud services; no server purchase is planned. Baseline hosting and API usage target free tiers, but service limits must be reviewed before the live demonstration. The existing Ubuntu VPS is not part of the baseline and remains only a possible contingency.

4 Project Plan

4.1 Process and Phase Gates

The project follows the course Waterfall process with V-model test planning. Each submitted PA becomes a baseline; subsequent changes are documented and reported in the following assignment. Preparatory implementation setup may overlap PA4 testing-plan preparation when it uses the PA3 design baseline.

Gate	Phase	Due	Exit criteria
PA1	Initiation and planning	End W3	Reviewed vision, proposal, estimates, schedule, risk and tool baseline
PA2	Requirements	End W5	Reviewed SRS, use cases, mockups and traceability
PA3	Software design	End W7	Approved architecture, UML, data and UI designs; ADRs
PA4	Test planning	End W8	Reviewed test strategy, techniques and Must-have test cases
PA5	Implementation / test / deployment	End W10	Deployed Must-have flows, recorded tests, CI results and release evidence
Final	Presentation and demo	W11	Demonstration and presentation materials ready

4.2 Work Breakdown Structure (WBS)

Level 1 groups the project into PA phases and continuous work. Level 2 groups deliverables; each three-level work package has one accountable owner and is suitable for estimation and Jira tracking. Owner responsibility does not prevent assistance by other members.

PA1 Initiation & Planning

WBS ID	Deliverable / Work package	Owner
1.1	Vision & Proposed Solution	–
1.1.1	Review vision, stakeholders and MVP scope	Ngoc
1.1.2	C4 context and preliminary solution	Quan
1.2	Management Baseline	–
1.2.1	WBS, estimates, schedule and cost plan	Hung
1.2.2	Risk register and tool setup	Tri
1.2.3	Review, consolidate and submit PA1	Hung

PA2 Requirements

WBS ID	Deliverable / Work package	Owner
2.1	Requirements Specification	–
2.1.1	Functional requirements and domain rules	Ngoc
2.1.2	NFRs and acceptance criteria	Ngoc
2.2	Behaviour Modelling	–
2.2.1	Use-case diagram and specifications	Ngoc
2.2.2	Core workflow mockups	Quan
2.3	Requirements Validation	–
2.3.1	Requirements review and traceability	Tri

PA3 Software Design

WBS ID	Deliverable / Work package	Owner
3.1	Architecture	–
3.1.1	C4 architecture and deployment diagrams	Quan
3.1.2	Architecture Decision Records	Huy
3.2	Data & Security Design	–
3.2.1	ERD, schema, constraints, indexes	Huy
3.2.2	Authorization and RLS design	Huy
3.3	Application Design	–
3.3.1	UI flows and key screens	Quan
3.3.2	UML class, sequence and state diagrams	Quan
3.4	Design Review	–
3.4.1	Design consistency review	Hung

PA4 Test Planning

WBS ID	Deliverable / Work package	Owner
4.1	Test Strategy & Design	–
4.1.1	Test strategy, environment and exit criteria	Tri

WBS ID	Deliverable / Work package	Owner
4.1.2	EP, BVA and decision-table design	Tri
4.2	Test Cases & Traceability	–
4.2.1	Must-have test-case specifications	Tri
4.2.2	Test coverage and traceability review	Ngoc

PA5 Implementation, Testing & Deployment

WBS ID	Deliverable / Work package	Owner
5.1	Implementation Foundation	–
5.1.1	Supabase schema, Auth and RLS	Huy
5.1.2	React app shell and authentication UI	Quan
5.1.3	Basic CI checks and test fixtures	Tri
5.2	MVP Feature Delivery	–
5.2.1	F01 cost/material management	Huy
5.2.2	F02 editable stage-planning UI	Quan
5.2.3	F02 Gemini planning Edge Function	Hung
5.2.4	F03 attendance and wage calculation	Huy
5.2.5	F04 context-aware AI chat UI	Quan
5.2.6	F04 contextual AI Edge Function	Hung
5.3	Verification & Release	–
5.3.1	Automated tests for selected use cases	Tri
5.3.2	Test execution and defect verification	Tri
5.3.3	Production deployment and smoke tests	Huy
5.3.4	PA5 report, evidence, retrospective	Hung

Final Presentation

WBS ID	Deliverable / Work package	Owner
6.1	Presentation & Demonstration	–
6.1.1	Presentation and demo script	Hung
6.1.2	Demo rehearsal and fallback plan	Quan

Continuous Project Management

WBS ID	Deliverable / Work package	Owner
7.1	Project Monitoring & Collaboration	–
7.1.1	Weekly meetings and status reports	Hung
7.1.2	Jira maintenance and contribution evidence	Hung
7.1.3	Repository standards and PR workflow	Huy

4.3 Estimation

Estimation method: bottom-up expert judgment, informed by the defined scope, likely implementation complexity and team capacity. Each work package estimate includes development/documentation work and local verification. Estimates are planned person-hours rather than elapsed duration; assistance by non-owner contributors is included in the work-package effort total.

PA1 Initiation & Planning

WBS ID	Work package	Owner	Effort	Method / assumption
1.1.1	Review vision, stakeholders and MVP scope	Ngoc	6 h	EJ; reuse PA0, align BA scope
1.1.2	C4 context and preliminary solution	Quan	7 h	EJ; C4 context only (PA3 later)
1.2.1	WBS, estimates, schedule and cost plan	Hung	9 h	EJ; 39 WPs, estimates + CPM
1.2.2	Risk register and tool setup	Tri	5 h	EJ; 8 risks and setup review
1.2.3	Review, consolidate and submit PA1	Hung	8 h	EJ; cross-review + DOCX/PDF
	PHASE TOTAL		35 h	

PA2 Requirements

WBS ID	Work package	Owner	Effort	Method / assumption
2.1.1	Functional requirements and domain rules	Ngoc	12 h	EJ; F01–F04 rules and flows
2.1.2	NFRs and acceptance criteria	Ngoc	8 h	EJ; measurable NFRs and checks
2.2.1	Use-case diagram and specifications	Ngoc	14 h	EJ; main and exception flows
2.2.2	Core workflow mockups	Quan	14 h	EJ; homeowner/crew MVP screens
2.3.1	Requirements review and traceability	Tri	7 h	EJ; requirements traceability
	PHASE TOTAL		55 h	

PA3 Software Design

WBS ID	Work package	Owner	Effort	Method / assumption
3.1.1	C4 architecture and deployment diagrams	Quan	9 h	EJ; C4 views building on PA1
3.1.2	Architecture Decision Records	Huy	6 h	EJ; ≥3 concise BaaS ADRs
3.2.1	ERD, schema, constraints, indexes	Huy	11 h	EJ; core relational entities
3.2.2	Authorization and RLS design	Huy	8 h	EJ; owner/crew RLS access rules
3.3.1	UI flows and key screens	Quan	12 h	EJ; UI flows and key screens
3.3.2	UML class, sequence and state diagrams	Quan	12 h	EJ; class, sequence, state UML
3.4.1	Design consistency review	Hung	6 h	EJ; PA2-to-PA3 design review
	PHASE TOTAL		64 h	

PA4 Test Planning

WBS ID	Work package	Owner	Effort	Method / assumption
4.1.1	Test strategy, environment and exit criteria	Tri	6 h	EJ; test levels, exit criteria
4.1.2	EP, BVA and decision-table design	Tri	8 h	EJ; EP/BVA + decision tables
4.2.1	Must-have test-case specifications	Tri	12 h	EJ; F01–F04 scenario coverage
4.2.2	Test coverage and traceability review	Ngoc	5 h	EJ; Must-have traceability
	PHASE TOTAL		31 h	

PA5 Implementation, Testing & Deployment

WBS ID	Work package	Owner	Effort	Method / assumption
5.1.1	Supabase schema, Auth and RLS	Huy	12 h	EJ; schema/Auth/RLS + access
5.1.2	React app shell and authentication UI	Quan	12 h	EJ; routing, shell and sign-in
5.1.3	Basic CI checks and test fixtures	Tri	7 h	EJ; lint/build/tests + fixtures
5.2.1	F01 cost/material management	Huy	16 h	EJ; CRUD, budget totals, charts
5.2.2	F02 editable stage-planning UI	Quan	12 h	EJ; stage CRUD and edit states
5.2.3	F02 Gemini planning Edge Function	Hung	11 h	EJ; Gemini draft + JSON checks
5.2.4	F03 attendance and wage calculation	Huy	17 h	EJ; attendance and wage math
5.2.5	F04 context-aware AI chat UI	Quan	11 h	EJ; chat UI and request states
5.2.6	F04 contextual AI Edge Function	Hung	11 h	EJ; bounded context + safeguards
5.3.1	Automated tests for selected use cases	Tri	14 h	EJ; 2 cases × 2 CI scenarios
5.3.2	Test execution and defect verification	Tri	13 h	EJ; defects and regression
5.3.3	Production deployment and smoke tests	Huy	7 h	EJ; Vercel/Supabase smoke test
5.3.4	PA5 report, evidence, retrospective	Hung	9 h	EJ; evidence and retrospective
	PHASE TOTAL		152 h	

Final Presentation

WBS ID	Work package	Owner	Effort	Method / assumption
6.1.1	Presentation and demo script	Hung	7 h	EJ; slides and demo narrative
6.1.2	Demo rehearsal and fallback plan	Quan	4 h	EJ; rehearsal and fallback
	PHASE TOTAL		11 h	

Continuous Project Management

WBS ID	Work package	Owner	Effort	Method / assumption
7.1.1	Weekly meetings and status reports	Hung	18 h	EJ; W2–W10 status reports
7.1.2	Jira maintenance and contribution evidence	Hung	12 h	EJ; Jira WBS/status upkeep
7.1.3	Repository standards and PR workflow	Huy	12 h	EJ; PR review and Git upkeep
	PHASE TOTAL		42 h	

Effort assumptions: PA2 focuses on four Must-have user journeys; PA3 designs managed Supabase components rather than a standalone backend; PA5 includes only F01–F04 and selective automation. Uncertain AI integration and access-control efforts will be re-estimated after prototyping and as technical evidence emerges.

Phase / activity	Person-hours
PA1 Initiation & Planning	35 h
PA2 Requirements	55 h
PA3 Software Design	64 h
PA4 Test Planning	31 h
PA5 Implementation, Testing & Deployment	152 h

Phase / activity	Person-hours
Final Presentation	11 h
Continuous Project Management	42 h
PROJECT TOTAL	390 h

4.4 Schedule and Critical Path

The schedule uses relative course weeks rather than guessed calendar dates. The two team-internal deadlines from PA0 apply to each tracked work package. Intended completion precedes peer review; the final internal deadline protects dependent tasks and the fixed submission gate. Review buffer size is adaptive to task complexity, risk and dependencies.

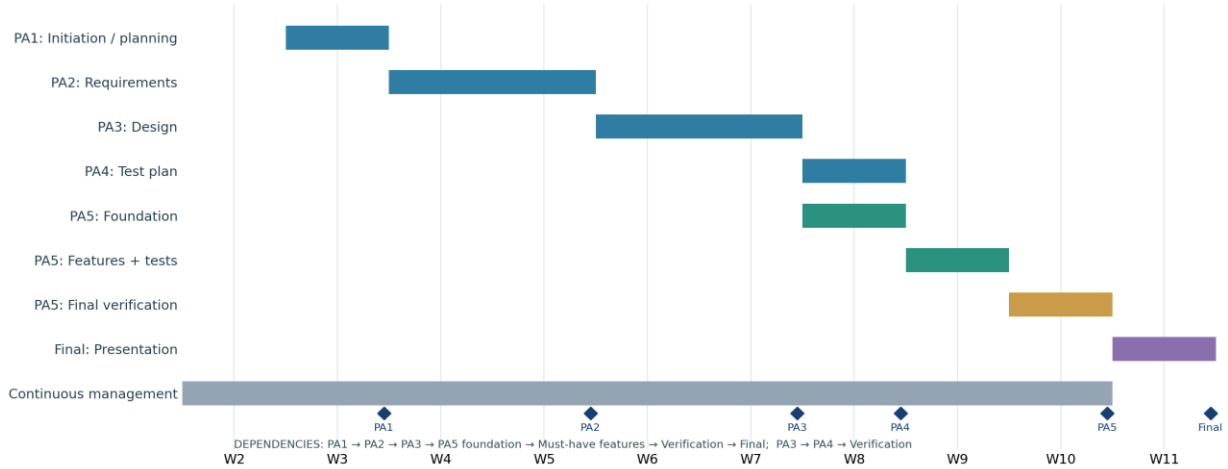


Figure 4.1 – Proposed Gantt (weeks W2–W11), PA gates and summary dependencies.

PA milestone	Intended completion	Final internal deadline	Official gate
PA1	W3: ~2 days before gate	W3: ~1 day before	End W3
PA2	W5: ~3 days before gate	W5: ~1 day before	End W5
PA3	W7: ~3 days before gate	W7: ~1 day before	End W7
PA4	W8: ~2 days before gate	W8: ~1 day before	End W8
PA5	W10: ~4 days before gate	W10: ~2 days before	End W10
Final	W11: ~2 days before demo	W11: ~1 day before	W11

Milestone buffers are planning targets, not achieved dates; Jira tracks package-level dates and dependencies. In W8, PA4 planning proceeds alongside PA5 foundation tasks on the PA3 baseline, including UI scaffolding, data contracts and mocked checks. W9 delivers tested Must-have slices; W10 prioritizes integration, defect fixes, regression and release. Official W8 and W10 gates remain fixed.

Dependency analysis and critical path (summary CPM)

To calculate a verifiable critical path at the summary-activity level, the following network models each phase-deliverable block in working days. Durations include internal review allowances; they are not obtained by equating person-hours to days. Dependencies are finish-to-start unless explicitly parallel. This is a planning network: finer-grained package dates will be maintained on Jira.

ID	Summary activity	Days	Pred.	ES	EF	Float
A	PA1 baseline	5	None	0	5	0
B	PA2 requirements	10	A	5	15	0
C	PA3 design	10	B	15	25	0

ID	Summary activity	Days	Pred.	ES	EF	Float
D	PA4 test plan	5	C	25	30	5
E	PA5 foundation	5	C	25	30	0
F	MVP implementation	5	E	30	35	0
G	Integration & release	5	D,F	35	40	0
H	Final demonstration	5	G	40	45	0

Summary CPM critical path: A → B → C → E → F → G → H = 45 working days (zero network float). PA4 test planning (D) has five days of network float relative to G, but none relative to its mandatory end-W8 submission gate; its internal review deadline still applies. The phase-level CPM simplifies a work plan in which implementation and verification overlap.

Dependencies: PA2 requirements → PA3 design; database/Auth/RLS foundation → integrated F01/F03; agreed data contracts → parallel UI and Edge Function work. Incremental testing starts as features become available. Final release requires the PA4 test baseline and completed Must-have flows. Figure 4.1 summarizes these predecessor links.

4.5 Roles and Responsibilities

Role	Main	Backup	Responsibilities
Team Lead / PM	Huỳnh Minh Hùng	Võ Thúc Trí	Planning, task assignment, weekly reporting, risk follow-up, integration of submissions.
Business Analyst	Nguyễn Phạm Minh Ngọc	Trần Gia Huy	Requirements, SRS, use cases, review and traceability.
Designer / Architect	Phạm Minh Quân	Huỳnh Minh Hùng	C4 and UML design, data/UI design, key screen flows and design documentation.
Developer	Trần Gia Huy	Phạm Minh Quân	Database integration, implementation, unit checks and pull-request reviews.
Tester	Võ Thúc Trí	Nguyễn Phạm Minh Ngọc	Test planning, test cases, test execution, defect tracking and test reporting.

The table records primary and backup roles from PA0; specific WBS ownership reflects the PA1 work plan. Additional members may contribute to work packages while the named owner remains accountable.

4.6 Human Resources and Cost Estimate

The hourly labour rate is a planning assumption, not salaries actually paid to students: 80,000 VND/person-hour, or 12.8 million VND per 160-hour work month. Economic labour cost is estimated from total planned effort; direct cash spending is separately estimated from expected service usage.

Activity	Effort	Rate (VND/h)	Labour cost (VND)
PA1 Initiation & Planning	35	80,000	2,800,000
PA2 Requirements	55	80,000	4,400,000
PA3 Software Design	64	80,000	5,120,000
PA4 Test Planning	31	80,000	2,480,000
PA5 Implementation, Testing & Deployment	152	80,000	12,160,000
Final Presentation	11	80,000	880,000
Continuous Project Management	42	80,000	3,360,000
TOTAL	390	80,000	31,200,000

Tools / infrastructure	Budgeted cash cost	Assumption
Vercel frontend	0 VND	Non-commercial student demonstration within eligible free-tier limits.
Supabase Cloud	0 VND	Managed platform stays within free allocation.

Tools / infrastructure	Budgeted cash cost	Assumption
Gemini API	0 VND	Usage monitored to remain within an applicable no-cost quota.
GitHub, Jira, Microsoft Teams	0 VND additional	Available free, institutional or already provisioned access.
Domain	0 VND	Uses default hosting-provided subdomain.
Optional contingency reserve	500,000 VND	Not expected expenditure; held for unforeseen quota or hosting charges.

Baseline estimated economic cost: 31,200,000 VND; planned reserve: 500,000 VND; total including reserve: 31,700,000 VND. Expected direct infrastructure outlay is 0 VND under these explicit service-tier assumptions, and may change if usage limits or pricing change.

5 Risk Management

Risks are classified into Technology, People, Organisational, Requirements and Estimation groups. The Project Management lecture (slides 38–41) defines RE = Probability × Loss and maps categorical ratings to numeric values. Here the team uses those values to prioritize potential adverse events. Ratings are preliminary judgments and are reassessed during weekly reporting.

Probability	Value	Loss	Value
Unlikely	10	Negligible	10
Seldom	25	Insignificant	25
Occasional	50	Moderate	50
Likely	75	Serious	75
Frequently	90	Catastrophic	90

ID	Risk	Type	P	L	RE	Owner
R01	Key member unavailable or delayed handoff	People	50	75	3,750	Hùng
R02	Implementation effort underestimated	Estimation	75	75	5,625	Hùng
R03	Unsafe or inaccurate AI technical guidance	Technology	75	90	6,750	Ngọc
R04	Incorrect authorization exposes private project data	Technology	50	90	4,500	Huy
R05	Supabase/Gemini unavailability or quota limits	Technology	50	75	3,750	Hùng
R06	Requirements ambiguity or scope creep	Requirements	75	75	5,625	Ngọc
R07	Delayed reviews and dependent handoffs	Organisational	50	75	3,750	Trí
R08	Critical defects found near PA5 release	Technology	75	75	5,625	Huy

5.1 Mitigation and Contingency Register

ID / owner	Preventive mitigation	Contingency if the risk occurs
R01 / Huỳnh Minh Hùng	Track both internal deadlines; publish shared work; maintain backup roles.	Reassign to a backup or redistribute tasks, replan dependent work.
R02 / Huỳnh Minh Hùng	Bottom-up estimates; compare actual vs planned work; prioritize F01–F04.	Rebalance staffing and defer optional scope while protecting Must-have criteria.
R03 / Nguyễn Phạm Minh Ngọc	Limit AI to advisory output; disclaimers; validate drafts; require owner confirmation.	Restrict unsafe responses, preserve manual editing and direct users to professionals.
R04 / Trần Gia Huy	Project-level Supabase RLS; server-side secrets; explicit cross-role permission tests.	Disable affected access, repair policies, retest before deployment.

ID / owner	Preventive mitigation	Contingency if the risk occurs
R05 / Huỳnh Minh Hùng	Early Edge Function integration; usage caps; bounded context; mock AI tests.	Use manual workflows temporarily; lower calls or change compatible AI model.
R06 / Nguyễn Phạm Minh Ngọc	MoSCoW baseline, numbered FRs, acceptance criteria and documented reviews.	Assess change impacts; defer nonessential changes; record baseline updates.
R07 / Võ Thúc Trí	Adaptive buffers, peer review, Jira blockers and weekly status follow-up.	Escalate to lead, change reviewers and reprioritize dependent tasks.
R08 / Trần Gia Huy	Prepare PA4 cases, continuous tests, and review code plus RLS-sensitive flows.	Fix blocking defects first; postpone optional work; rerun regressions and smoke tests.

5.2 Risk Monitoring

Risk owners review probability, loss, exposure and early indicators in the weekly project status meeting. Indicators include missed intended/final deadlines, unexpected changes in scope, repeated API errors, authorization-test failures, unavailable members and unresolved high-severity defects. Materialized risks trigger the recorded contingency action and immediate escalation to the Team Lead; high-exposure risks are also noted in weekly status reports.

RE provides relative prioritization, not a forecast of monetary loss. In this baseline, unsafe AI technical guidance (R03; RE 6,750), underestimated effort (R02; 5,625), scope ambiguity (R06; 5,625) and release defects (R08; 5,625) warrant particular attention.

6 Tools & Collaboration Setup

The team established GitHub, Jira and Microsoft Teams during PA0. Setup is partially complete at PA1 drafting time; the document distinguishes resources already created from policies or configurations that remain to be verified. The task board and repository will be aligned with the PA1 WBS before the relevant grading checkpoint.

Tool	Purpose	PA1 setup status
Moodle	Official announcements and PA submissions	Required submission platform
Microsoft Teams	NMCNPM / Nhóm 8 / General; team and TA communication	Channel established; confirm TA membership
Jira	WBS tasks, owners, due dates, statuses and dependencies	Board established; import/verify WBS tasks and deadlines
GitHub	Source, documentation, PR review and deliverables	Repository established; verify branch protection and teaching-staff access
Google Drive	Supporting shared drafting and file exchange	Shared workspace listed in PA0

6.1 Communication and Working Cadence

The primary communication channel is Microsoft Teams (NMCNPM / Nhóm 8 / General). The internal meeting is Sunday at 09:30 and the online TA status meeting is Sunday at 19:30. Members aim to acknowledge ordinary messages within 24 hours, with escalation for urgent blockers. Decisions and follow-ups are recorded on Jira or in meeting notes.

Weekly status reports are due Sunday at 00:00 according to the PA0 team contract. One immutable report per week from W2 to W10 is committed to /docs/management/weekly/W<week>-status.docx (or .md). The Team Lead or backup is responsible. Existing reports are not rewritten.

6.2 Task Board and Deadline Policy

Jira holds one trackable record per WBS work package, with accountable owner, intended completion date, final internal deadline, status and dependency information. The review buffer between deadlines is adaptive rather than fixed. When either deadline is at risk, the owner flags the blocker and expected impact promptly.

6.3 Repository and Review Policy

- Work is prepared on feature/task branches and integrated into main via peer-reviewed pull requests (at least one reviewer).

- Main-branch protection, teaching-staff access and related repository settings are agreed requirements; their activation must be verified rather than assumed.
- A README and .gitignore describe setup and exclude local credentials; .env files and API keys must never be committed.
- Commit messages follow Conventional Commits (e.g., feat:, fix:, docs:, test:).
- Each PA is committed as PDF and editable source under /pa/PAx; official submission also occurs in Moodle.

6.4 Expected Repository Structure

Path	Content
/src	Application source code
/docs/management	Project plan and administrative records
/docs/management/weekly	Weekly status reports W2–W10
/docs/requirements	Vision, SRS, use cases and mockups
/docs/design	C4/architecture decisions, UML and UI design
/docs/test	Test plan, cases and reports
/docs/ai-usage	AI usage logs and supporting materials
/pa/PA1 … /pa/PA5	Assignment PDFs and editable source files

Evidence still to be checked before final submission: protected main branch, at least one review on pull requests, instructor/TA access, sufficient Jira issues with owners/due dates, and repository folders. The proposal describes the operating policy without claiming those verifications are complete.

7 AI Usage Log

The following records disclose significant AI-assisted work during PA1. All submitted material remained subject to human review, correction and team approval. The exact AI tool names used by the Business Analyst and Developer require confirmation before submission.

#	Member	AI tool	Task / purpose	Verification and changes
1	Nguyễn Phạm Minh Ngọc	Google Gemini	Assisted drafting Section 2: problem statement, product positioning, stakeholders, and feature scope.	Discussed and self-reviewed the AI-assisted draft. The team lead independently reviewed it, made refinements, and retained the largely accepted business content.
2	Trần Gia Huy	Claude	Assisted drafting Section 3: preliminary architecture, technology stack, and operating environment.	Discussed and self-reviewed the proposal. After team-lead review, the over-complex dedicated backend/deployment approach was replaced with Supabase BaaS and Edge Functions; the revision was agreed with the Developer.
3	Huỳnh Minh Hùng	ChatGPT	Translated, consolidated and refined the BA and architecture drafts into the English PA1 report; discussed solution alternatives.	Checked changes against PA0, the course template, project scope and the team agreement. Personally selected the simplified architecture and reviewed the integrated sections.
4	Huỳnh Minh Hùng	ChatGPT	Assisted with WBS, estimation, Gantt/CPM, costs, risk register, collaboration text, document assembly, and QA.	Reviewed assumptions and revised drafts; checked the 390-hour and cost totals, matched RE scores to the lecture scale, and inspected the Word layout and schedules.